

2.2 Vanishing and Exploding Gradients

Backprop through time is a product of Jacobians. Long-range signal dies or blows up.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

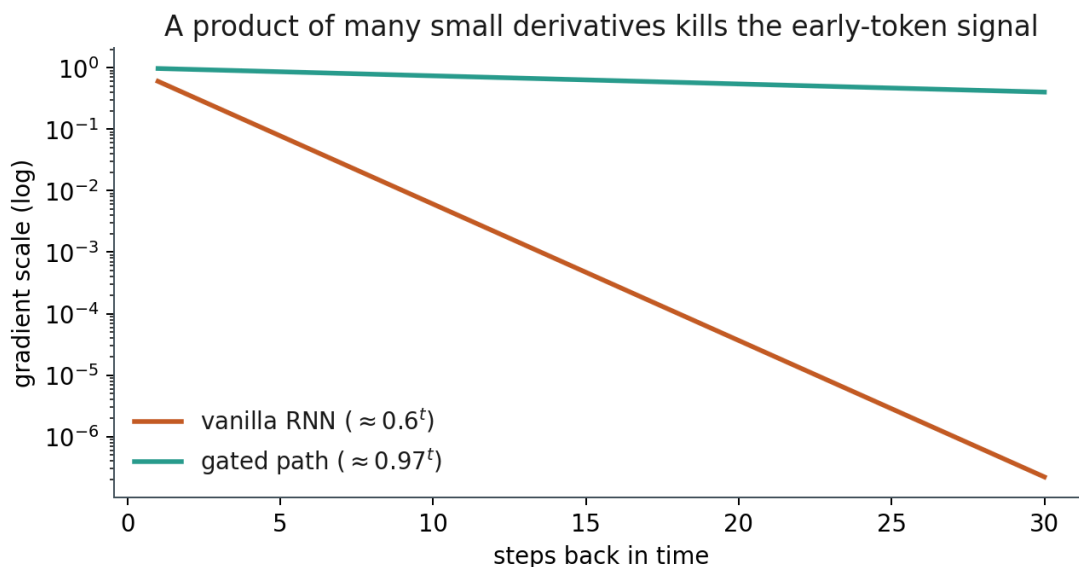
Backprop through an unrolled RNN multiplies one Jacobian per time step. If those matrices shrink, the early-token gradient **vanishes**. If they grow, it **explodes**. Either way, long-range learning dies.

1. A product along the chain

The loss at the last token, $L(h_T)$, depends on h_1 through

$$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial h_T} \frac{\partial h_T}{\partial h_{T-1}} \cdots \frac{\partial h_2}{\partial h_1}.$$

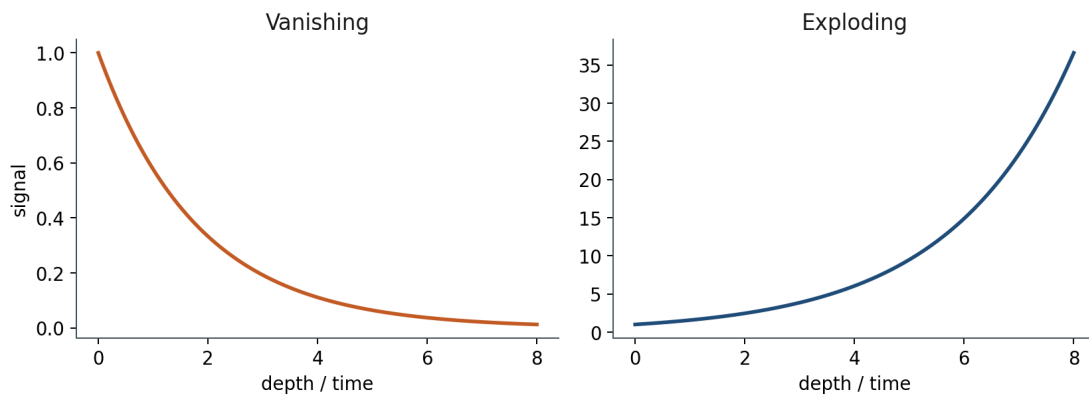
Each factor $\partial h_t / \partial h_{t-1}$ contains W_h and $\tanh'$. For $\tanh$, $|\tanh'| \leq 1$, and it is $\ll 1$ unless the pre-activation sits near 0. A product of twenty numbers smaller than 1 is tiny.



The gated curve is the slogan for LSTM/GRU: keep a path whose multiplier can stay near 1.

2. Vanishing versus exploding

Same product, opposite problem. If $\|W_h\|$ is large, the signal blows up, weights become NaNs, and training stops. **Gradient clipping** (rescale if the global norm exceeds a threshold) is the usual bandage for exploding. It does not fix vanishing.



3. What this means for language

A vanilla RNN can learn “the next word looks like the last few.” It struggles with “the negation was at the start of the review.” Transformers side-step the chain: every token attends to every other token in one step (Week 3). Gates (Week 2.3) were the fix *inside* the recurrent family.

In a project, if you still train an RNN on long documents, watch the gradient norm. If it is always ≈ 0 on early positions, the net is not using the beginning of the sequence.

4. Practice

1. Why is $\tanh'$ a problem when the state saturates at ± 1 ?
2. Clipping stops explosions. Why does it not restore a vanished signal?
3. In one sentence: what would you plot in TensorBoard to see this failure in Lab 2?